

Knowledge before Industrialization:

Evidence from Early Copyright Records

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Abstract

This paper uses new data on copyright registration title pages from the Library of Congress to study intellectual and cultural production in the United States over 1790–1870. I construct national series that show a sustained rise in copyright registrations per capita beginning around 1830, before the Second Industrial Revolution. Linking authors to the 1850 census locates registrations from 1840–1860 at the county level. Counties with more copyright activity subsequently recorded higher patenting rates over 1860–1940 and larger manufacturing employment shares during industrialization. The empirical strategy uses a shift-share-type instrument based on differences in authorship rates across origins and the settlement patterns of migrants.

Keywords: Knowledge production; copyright; human capital; industrialization; U.S. economic history.

JEL classification: N31, N61, O14, O31.

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1 Introduction

The accumulation and application of knowledge lie at the heart of sustained economic growth (Mokyr, 2002). New ideas can improve production, while their preservation and circulation allow further advances to build on what is already known (Mokyr, 2009, 2016). Understanding the origins of industrialization therefore requires examining the intellectual activity behind technological change: where knowledge was produced, what subjects it addressed, and how its development related to the transformation of the economy.

Yet knowledge is difficult to observe in historical data. Schooling and literacy describe aspects of people’s capacity to acquire and use knowledge, while patents record particular inventions. These measures tell us less about the broader range of written work through which ideas were articulated and communicated, or about how its subject matter changed. Copyright records provide a way to study this intellectual activity: they identify the works people produced and the subjects they addressed. Moreover, studies linking growth to knowledge production suffer an obvious endogeneity problem: knowledge may plausibly produce growth but the reverse is also almost certainly true (Baten and van Zanden, 2008).

This paper uses new data on copyright registration title pages from the Library of Congress to examine the production and composition of written knowledge and their relationship with American industrial development. These records cover copyright registrations preserved in the Library of Congress collection for 1790–1870 and allow me to construct national series of written production and study the composition and distribution of knowledge before the industrial expansion of the late nineteenth century. I link authors of works registered in 1840–1860 to the 1850 population census, obtaining a county-level measure of knowledge production at the outset of this expansion. I then ask whether counties with more copyright activity subsequently developed larger manufacturing employment shares and higher patenting rates. By classifying the copyright works into knowledge categories, these records also allow me to examine whether the manufacturing relationship is confined to academic writing or extends to other forms of written production.

Three findings organize the analysis. First, at the national level I document a marked expansion of copyright activity during the first half of the nineteenth century, well before the start of the so-called era of “knowledge-based progress” (Abramovitz and David, 2000). Registrations per capita begin a sustained rise around 1830, preceding the sharper acceleration in patenting visible in the following decades. This increase took place before the industrial expansion of the late nineteenth century.

Second, written knowledge production had a distinct local geography. The Northeast was the principal center of copyright activity, with substantial variation across counties within the same state. Authors were concentrated in larger, more urban, and more literate counties with a stronger educational presence. The census links make it possible to locate these producers of written work and to study their contribution to the geography of human capital in the early United States. The county analysis builds on this variation, comparing places within states while accounting for geography, population, migration, and conventional measures of education.

Third, this early geography of intellectual production predicts subsequent manufacturing and invention. Counties with more copyright activity developed larger manufacturing employment shares during the industrial expansion, with the strongest OLS relationship around the early

twentieth century. Early copyrighting also predicts patenting over 1860–1940 after accounting for a rich set of geographic, educational, and social controls. These findings show that written production contains information about industrial development beyond that captured by literacy, schooling, and the initial scale of local economic activity. The association with manufacturing follows the historical rise of the sector and recedes later in the twentieth century, while the relationship with population density persists.

To sharpen identification, I exploit the relationship between internal migration and the location of authors. I construct a shift-share-type instrument by combining county birthplace shares with authorship rates among people from each origin. Counties receiving more migrants from origins with a greater propensity for authorship have higher predicted copyright activity. The instrument uses out-of-state origins and selected foreign birthplaces, drawing on the role of migrant networks in settlement patterns (Bartel, 1989; Munshi, 2003). Identification requires that, conditional on the included controls, the origin composition of settlers affect later outcomes through local knowledge production. I examine geographic balance and introduce controls sequentially to distinguish this variation from differences in education and settlement. The IV estimates yield positive relationships with manufacturing and patenting in the specifications controlling for geography, human capital, and baseline urbanization.

The paper contributes to research on knowledge and historical growth (Mokyr, 2005, 2009, 2016) by bringing the scale, subject matter, and location of written production into a common empirical setting. Its closest connection is to Squicciarini and Voigtländer (2015), who use subscriptions to the *Encyclopédie* to study upper-tail human capital and industrialization in France. The copyright records identify the producers of written work, and the census links recover their residences and birthplaces. This combination extends the use of books and printing to study historical development (Baten and van Zanden, 2008; Dittmar, 2011) and connects early intellectual production to the subsequent geography of American invention (Sokoloff, 1988; Khan and Sokoloff, 1993).

The county evidence also contributes to studies of migration and American development (Sequeira et al., 2020) and of the historical sources of structural change (Fiszbein, 2022). It shows that the presence of authors offers a distinct perspective on local human capital during the industrial transition. The composition results further establish that the manufacturing relationship extends to non-academic writing, connecting industrial development to a broad range of intellectual activity.

Section 2 describes the archive, census linkage, and measurement. Section 3 presents the national and county evidence. Section 4 develops the empirical specification, migration instrument, and identifying assumptions. Section 5 reports the main results, and Section 6 concludes. The appendices document matching diagnostics, variable definitions, classification, and supplementary results.

2 Copyright records and census linkage

2.1 The copyright archive

The dataset measures written production through copyright title pages preserved by the Library of Congress (Library of Congress, n.d.). The records identify works submitted for protection

and provide information on their titles, authors, dates, and places of publication. During the period studied, applicants filed title pages with federal district courts. In 1870, responsibility for registration was centralized at the Library of Congress, which also received surviving records from the earlier system (Khan, 2005).

The archival description estimates approximately 50,000 works represented in more than 90,000 images. The images include title pages, accompanying forms, and blank pages. After processing the transcriptions, I can assign dates to approximately 70,000 entries. A dated record is the unit of observation in the national series: renewals, resubmissions, and duplicate scans can generate several records for a single work. The series therefore traces the volume of recorded copyright activity and its evolution over time.

2.2 Extracting dates and author names

I collect the title-page transcriptions and extract dates and author names from their text. The dating procedure converts Roman numerals into Arabic numerals and identifies numerical sequences beginning with “17” or “18”. These dated entries form the basis of the annual and decadal registration series.

To identify authors, I use the text following the first occurrence of “by” and compare its words with first and last names in the 1850 census. For works with multiple authors, the procedure identifies the first author. The extracted names provide the link between the copyright archive and individual census records.

2.3 Linking authors to the 1850 census

The census locates authors of works registered between 1840 and 1860. All registrations in this window are assigned to the matched author’s county of residence in 1850. This common residential benchmark measures the written output associated with a mid-century county population, including works registered before and after the census.

Linkage proceeds in stages. First, I extract the publication state when it can be identified and use it to restrict the initial census search. The census record then supplies the author’s residential county. Second, I apply Soundex to the extracted names and census names to accommodate spelling variation in historical records. Third, I restrict candidates to the occupational groups listed in Appendix Table A2 and require an implied age of at least 25 at the time of the work. A second search extends to other states for names that remain unmatched.

The occupational and age restrictions define the group of writers represented in the linked sample. Appendix Table A1 reports the resulting candidate multiplicity: 71.69% of entries have a single eligible census candidate. This statistic measures uniqueness conditional on the name, state, occupation, and age criteria. Appendix A presents the detailed diagnostics, occupational composition, and author characteristics.

The most extensive OLS specifications include baseline population and related social controls. These account for differences in county size that influence both the volume of writing and the number of potential census matches.

2.4 County measures and other data

For each county, I aggregate the registrations from 1840–1860 assigned through the author links. The explanatory variable is log copyright registrations, denoted k_i . Authors with several registrations contribute correspondingly more to the county total. This measure captures the volume of recorded written output, while the birthplace instrument introduced below uses authorship rates among individuals.

County outcomes and many controls are drawn from the historical data assembled by Fiszbein (2022). Additional measures of urbanization, literacy, school enrollment, and religious denominations come from the historical census compilation of Haines and ICPSR (2010). National patent totals are from U.S. Patent and Trademark Office (2022). Appendix B describes the variables and sources used in the analysis.

The main outcomes are the manufacturing share of employment at selected dates, patents per capita per decade over 1860–1940, and the change in the urban population share between 1860 and 1940. Population density traces the longer-run concentration of economic activity, and income per capita in 2000 provides a supplementary outcome. Each table reports the sample available for its outcome and specification.

The regressions distinguish three groups of covariates. Geographic controls include terrain ruggedness, distance to the ocean or Great Lakes, climate, and land suitability. Human-capital controls comprise literacy and school enrollment. Social controls include population, population density, urbanization, and the shares of foreign-born and internal-migrant residents. Social and human-capital characteristics are measured in 1850. The sequential specifications add baseline urbanization separately before introducing the remaining social controls; the tables identify each control set.

3 The expansion and geography of copyrighting

3.1 National patterns, 1790–1870

Figure 1 shows the expansion of copyright activity between 1790 and 1870. Annual registrations rise substantially from their low levels in the earliest decades, with particularly strong growth around the middle of the nineteenth century. The decline in the early 1860s interrupts this longer-run expansion.

Copyrighting also expands relative to population. Figure 2 shows a sustained rise in registrations per capita from around 1830, followed by the pronounced increase in patenting in the subsequent decades. The timing places the expansion of recorded written production alongside early manufacturing growth and ahead of the larger industrial transformation of the late nineteenth century (Davis, 2004; Sokoloff, 1988). The county analysis builds on this national pattern by examining whether places with more early copyright activity subsequently experienced more manufacturing and invention.

3.2 County patterns, 1840–1860

Figure 3 shows the geographical concentration of copyright activity. Many of the highest concentrations lie in the Northeast, alongside substantial variation within states and activity

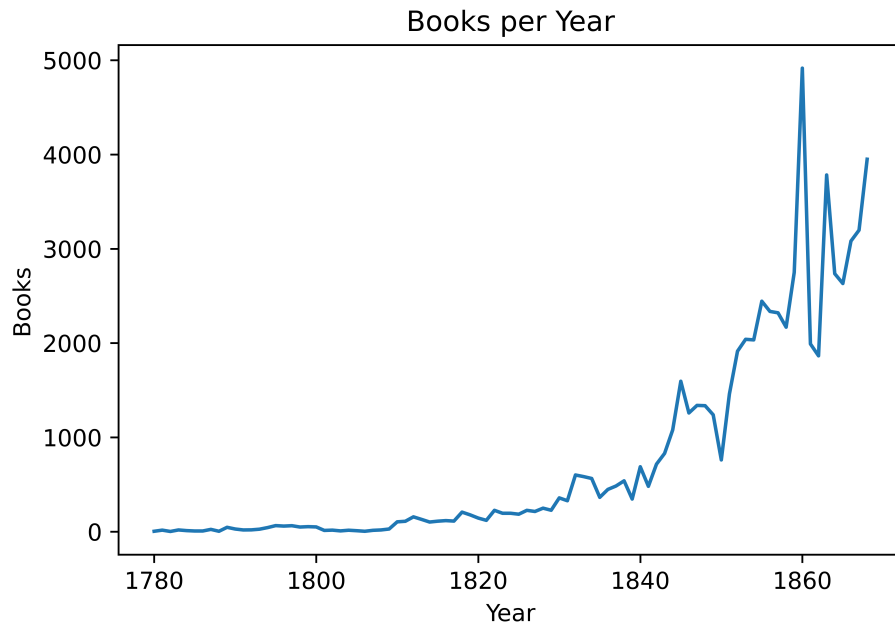


Figure 1. Annual copyright records, 1790–1870

Notes: Counts of dated records constructed from Library of Congress title-page transcriptions. Records can include renewals, resubmissions, and duplicate scans. Source: author’s calculations from Library of Congress (n.d.).

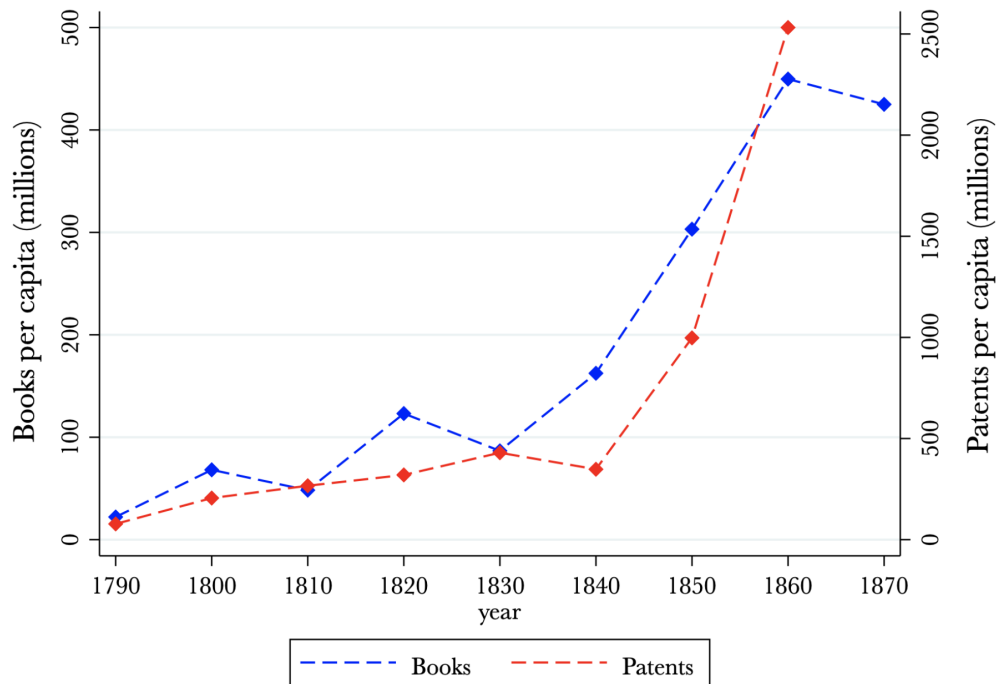


Figure 2. Copyrighting and patenting relative to population

Notes: Decadal population-normalized series. Copyrights use the left axis and patents the right axis; the axes have different scales. “Books” denotes dated copyright records. Sources: author’s calculations from Library of Congress (n.d.) and national patent statistics from U.S. Patent and Trademark Office (2022).

elsewhere in the country. The map distinguishes academic and non-academic works using title-page text. The academic category includes instructional as well as scientific writing; Appendix C describes the classification procedure. The main regressions use total registrations across both categories.

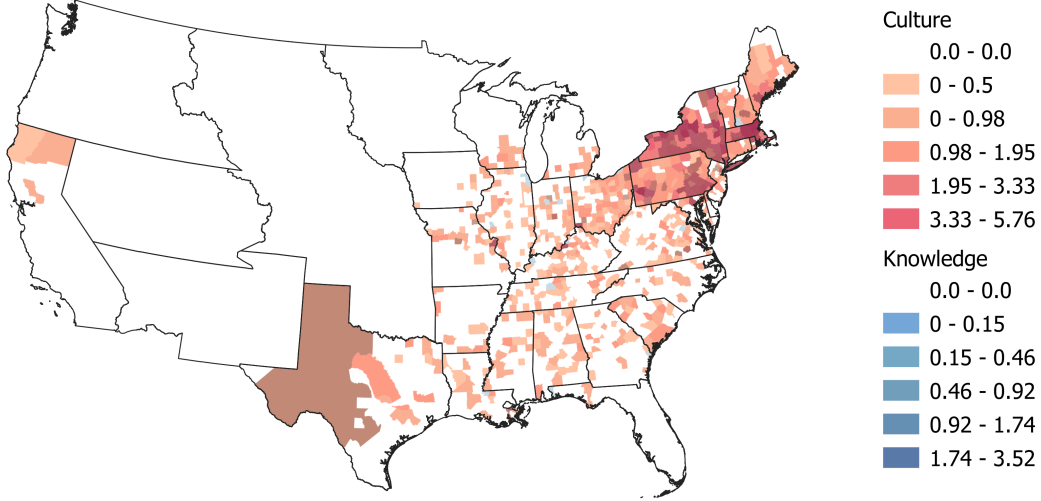


Figure 3. The county geography of linked copyright activity, 1840–1860

Notes: Registrations are assigned to the matched author’s county of residence in the 1850 census. The map uses “Knowledge” (blue) for works classified as academic and “Culture” (red) for non-academic works; overlapping colors appear purple. Shading displays log-count categories. Sources: author’s calculations from the copyright records and census links.

Table 1 compares counties above and below the median of copyright activity in the descriptive sample. Counties above the median have larger populations, higher literacy and urbanization, more schools per child, and more colleges. Written production was thus concentrated in places with greater population and educational provision. The empirical specifications account for these characteristics when relating early copyrighting to subsequent development.

Table 1. County characteristics by copyrighting intensity

	Above median		Below median		Difference <i>t</i> statistic
	Mean	SD	Mean	SD	
Population	25797.150	33419.058	8367.849	5061.875	−12.923
Literacy rate	0.876	0.114	0.792	0.151	−11.134
Urban population share	0.021	0.120	0.000	0.000	−4.284
Enslaved population share	0.167	0.223	0.211	0.226	3.433
Number of colleges	0.306	0.720	0.053	0.230	−8.390
Schools per child	0.020	0.014	0.015	0.012	−5.605
Counties	628		629		1257

Notes: Counties are divided at the median number of copyright registrations. The final column reports difference-in-means *t* statistics, with the below-median group first in the comparison. Literacy, urbanization, and the enslaved population share are fractions. Selected characteristics are shown. Sources: copyright records from the Library of Congress; county characteristics from Haines and ICPSR (2010).

The next section develops these comparisons. State fixed effects isolate within-state variation, baseline controls account for observed county characteristics, and the migration instrument uses differences in residents’ birthplaces to predict local authorship.

4 Empirical strategy

4.1 County comparisons

For county i in state $s(i)$ and outcome year t , the empirical specification is

$$y_{it} = \alpha_{s(i),t} + \beta_t k_i + X_i' \gamma_t + P_t(\ell_i, a_i) + \varepsilon_{it}, \quad (1)$$

where k_i is log copyright registrations from 1840–1860, X_i contains the county controls, and $P_t(\ell_i, a_i)$ is a third-degree polynomial in longitude and latitude. I estimate a separate county cross-section for each outcome year. For outcomes defined over an interval, such as patenting in 1860–1940, t denotes that period. State fixed effects focus the comparison on counties within the same state, while the spatial polynomial accounts for broad geographical trends.

The sequence of controls distinguishes written knowledge production from other features of the local economy. Geography accounts for natural endowments; literacy and school enrollment capture broader educational attainment; and urbanization, population, and migration describe the scale and composition of settlement. The main OLS tables introduce these groups successively. The selected-year manufacturing OLS estimates include the full set throughout. Inference allows spatial dependence using the [Conley \(1999\)](#) correction with a 100-kilometer cutoff. Parentheses in the tables report t statistics based on these standard errors.

Holding copyright activity fixed at 1840–1860 traces the relationship between mid-century written output and subsequent development. The 1850 manufacturing outcome provides a contemporaneous benchmark; later outcomes describe the industrial trajectory of counties with different initial levels of copyrighting. The baseline controls are also measured in 1850. Conditioning on them asks whether written activity predicts that trajectory beyond the educational and settlement characteristics already present at mid-century. The migration instrument develops a complementary comparison based on the origins of the people who settled in each county.

4.2 Migration and the construction of the instrument

Internal migration creates substantial variation in the origin composition of county populations. I define an internal migrant as a U.S.-born person whose state of residence in 1850 differs from the state of birth. Table 2 shows that authors were more likely to be internal migrants than non-authors in each reported region. In the combined sample, the respective shares are 36% and 24%. The contrast is particularly pronounced in the Midwest, where 78% of authors and 45% of non-authors lived outside their birth state.

Table 2. Interstate migration among native-born residents in 1850

Region	Share living outside state of birth		Observations
	Non-authors	Authors	
Midwest	0.45	0.78	4,584,383
Northeast	0.12	0.30	7,207,901
South	0.22	0.48	5,434,905
Total	0.24	0.36	17,227,189

Notes: An interstate migrant is a native-born individual residing in a different state from the state of birth in 1850. Observations give the size of the regional sample. Sources: 1850 Census of Population and Library of Congress copyright records.

The instrument combines this variation in settlement patterns with differences in the prevalence of authorship by birthplace. Let A_l be the number of identified authors born in origin l who registered a work during 1840–1860, and let N_l be the number of people in the 1850 U.S. census born in that origin. Define the origin-specific authorship rate and the corresponding national rate as

$$q_l = \frac{A_l}{N_l}, \quad q_{US} = \frac{A_{US}}{N_{US}}. \quad (2)$$

Let π_{il} denote the share of county i 's population born in origin l . The instrument is

$$Z_i = \sum_{l \in \mathcal{L}, l \neq s(i)} (q_l - q_{US}) \pi_{il}, \quad (3)$$

where $\mathcal{L}$ contains U.S. birth states and the United Kingdom, France, and Germany. For foreign origins, q_l measures authorship among U.S. residents born in the relevant country. The sum excludes residents born in the destination state, so the instrument draws on the county's connections to other origins.

Equation (3) translates the composition of settlement into an index of local authorship propensity. A county receives a higher value when more residents come from included origins with above-average authorship rates. Subtracting q_{US} centers each origin's rate on the national average. The rates count people who are authors, while the county regressor counts their registrations: the first stage measures how strongly the presence of populations with higher authorship propensities predicts local written output.

The first-stage equation is

$$k_i = \eta_{s(i)} + \rho Z_i + X_i' \delta + P(\ell_i, a_i) + v_i. \quad (4)$$

The same controls enter the second stage. The rationale follows the role of birthplace networks in migration: the settlement patterns of the broader population help predict the destinations of authors from the same origins. The design uses county birthplace shares observed in 1850 and authorship rates calculated from national census populations and linked authors active during 1840–1860. Identification therefore comes from the composition of settlement in this historical cross-section, under the conditional exclusion restriction set out below.

4.3 Identifying assumption and geographic balance

The identifying assumption is that, conditional on the included covariates, the settlement-composition index is orthogonal to unobserved determinants of later county outcomes. In the notation of equation (1),

$$\mathbb{E}[\tilde{Z}_i \varepsilon_{it}] = 0, \quad (5)$$

where $\tilde{Z}_i$ is the residual from projecting Z_i on the included state effects, controls, and spatial polynomial. Substantively, the assumption rules out an independent effect of origin composition on development through omitted characteristics of settlers or their destinations. The controls give this comparison a concrete historical basis: they account for geography, conventional human capital, and the observed structure of settlement. The geographic balance exercise examines

the spatial sorting underlying the instrument, and the sequence of social controls assesses the importance of population and migration composition.

The balance exercise in Table 3 regresses the instrument on geographic characteristics:

$$Z_i = \mu_{s(i)} + G_i'\theta + P(\ell_i, a_i) + u_i, \quad (6)$$

with state effects and the spatial polynomial added sequentially. This exercise establishes which observable geographical characteristics covary with the settlement-based prediction of authorship.

Table 3. Geographic correlates of the migration-based instrument

	(1)	(2)	(3)
Terrain ruggedness	0.811 (3.79)	0.848 (3.18)	0.581 (3.19)
Distance to ocean or Great Lakes (km)	−0.0367 (−9.03)	−0.0333 (−5.55)	−0.0102 (−1.43)
Temperature	−1.069 (−3.22)	0.947 (1.71)	1.514 (1.58)
Annual rainfall	0.000104 (0.02)	−0.00642 (−0.81)	0.00652 (1.24)
Land suitability	−1.744 (−0.50)	4.738 (1.73)	4.760 (1.89)
Counties	1,507	1,507	1,507
R^2	0.423	0.145	0.235
State fixed effects	No	Yes	Yes
Latitude–longitude polynomial	No	No	Yes

Notes: The dependent variable is the migration-based instrument. Column (3) includes a third-degree polynomial in latitude and longitude. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

With state effects and the spatial polynomial included, ruggedness is the only geographical characteristic significantly associated with the instrument at the 5% level. Its association is positive in all three columns. Distance to the ocean or Great Lakes is negatively associated with the index in the first two specifications, but the relationship is statistically insignificant after adding the spatial polynomial. The temperature coefficient changes sign and loses significance with state effects; rainfall and land suitability are insignificant at the 5% level throughout. Geography thus accounts for an observable component of settlement composition, including a persistent association with ruggedness. I retain these geographical controls in the main IV specifications and assess the additional role of settlement characteristics through the control sequence.

4.4 First-stage strength and the control sequence

The selected-year manufacturing IV specifications include geography, human capital, and baseline urbanization. Their Kleibergen–Paap statistics are 23 in 1850 and 13 in the subsequent columns. These specifications exclude the remaining social controls, whereas the manufacturing OLS estimates include them. Both the estimator and the control set therefore change between the two panels.

The urbanization, patenting, and income tables show the full control sequence. Adding population and migration controls reduces their first-stage statistics to approximately one

and produces highly imprecise IV estimates. These variables absorb much of the settlement-composition variation used by the instrument. The tables report the complete sequence, and the discussion of IV magnitudes focuses on specifications that retain a stronger first stage.

The empirical argument combines the fully controlled OLS relationships with the migration-based IV comparison. OLS establishes how copyrighting relates to development after accounting for observed county characteristics. The IV estimates examine the component of written activity predicted by the origins of local residents, under equation (5). Their interpretation rests jointly on that restriction and the first-stage evidence. Differences in their magnitudes can reflect measurement error as well as the change in controls and identifying variation; the historical patterns across outcomes provide the central basis for interpreting the results.

5 Copyrighting and subsequent development

5.1 Manufacturing employment and structural change

Counties with more copyright activity had larger manufacturing employment shares during the industrial expansion. Table 4 traces this relationship across selected years. With the full set of OLS controls, the copyright coefficient is positive in 1850, 1870, and 1920 and reaches its largest value in 1920. The comparison is within states and accounts for geography, literacy, schooling, and baseline population and settlement characteristics. Early authorship thus distinguishes counties with greater subsequent manufacturing specialization even after accounting for these other dimensions of local development.

Table 4. Copyrighting and the manufacturing employment share

Outcome year	1850	1870	1920	1970	2000
<i>Panel A. Ordinary least squares: full controls</i>					
Log copyright registrations	0.00742 (5.65)	0.00927 (3.57)	0.0111 (3.50)	−0.00324 (−1.06)	−0.00239 (−0.51)
Counties	1,114	1,504	1,504	1,504	1,504
R^2	0.271	0.193	0.204	0.221	0.129
Remaining social controls	Yes	Yes	Yes	Yes	Yes
<i>Panel B. Instrumental variables: excluding remaining social controls</i>					
Log copyright registrations	0.0630 (16.47)	0.0360 (0.44)	0.185 (3.88)	0.102 (1.96)	0.0350 (0.81)
Counties	1,114	1,505	1,505	1,505	1,505
Kleibergen–Paap F statistic	23	13	13	13	13
Remaining social controls	No	No	No	No	No
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geography controls	Yes	Yes	Yes	Yes	Yes
Human capital controls	Yes	Yes	Yes	Yes	Yes
Urbanization, 1850	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is manufacturing’s share of employment in the year shown. Copyright registrations refer to 1840–1860 and are assigned to authors’ counties of residence in 1850. Both panels include baseline urbanization, human capital and geography controls, state fixed effects, and the spatial polynomial. The OLS panel additionally includes population, population density, and the foreign-born and internal-migrant shares; these additional controls are excluded from the IV panel. The two panels therefore differ in controls and, after 1850, slightly in sample size. Only the copyright coefficient is displayed. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

The IV estimates reinforce the historical pattern. The migration-based specification yields positive coefficients in every selected year and, like OLS, its largest estimate in 1920. In that year, the OLS coefficient is 0.0111 with a t statistic of 3.50, and the IV coefficient is 0.185 with a t statistic of 3.88. The IV specification retains geography, human capital, baseline urbanization, and state and spatial controls; the OLS specification also includes the remaining population and migration controls. Their magnitudes therefore refer to different conditional comparisons. Because the dependent variable is an employment share measured as a fraction, the coefficients describe changes in the level of that share per unit change in log copyrighting.

Figure 4 brings out the timing of the relationship. The fully controlled OLS coefficient rises from 0.00742 in 1850 to 0.0111 in 1920, then becomes small and statistically indistinguishable from zero in 1970 and 2000. The IV estimates also fall after their 1920 peak. The concentration of the relationship around the period of industrial expansion is central to the interpretation: early copyright activity was associated with the places that became more specialized in manufacturing as the American industrial economy developed.

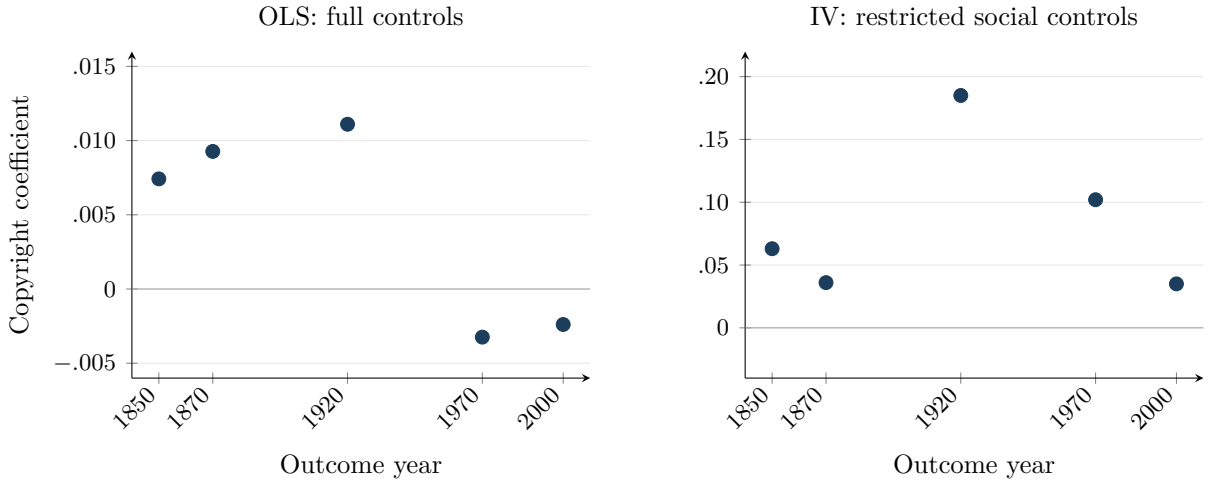


Figure 4. The changing relationship between copyrighting and manufacturing

Notes: Point estimates from Table 4, with separate vertical scales for OLS and IV. Manufacturing employment shares are measured as fractions. Both panels include geography, human capital, baseline urbanization, state effects, and the spatial polynomial. OLS additionally includes the remaining social controls. The 1850 outcome is contemporaneous with the registration window. Sample sizes, t statistics, and first-stage diagnostics are reported in the table.

The later decline in the manufacturing association gives the results a structural-change interpretation. Manufacturing’s employment share describes the sectoral composition of a county’s economy. Its evolution can therefore differ from that of the county’s population or overall economic scale. The population-density evidence below shows that the concentration of population in counties with more early copyright activity persisted beyond the period of their strongest association with manufacturing.

5.2 Inventive activity, 1860–1940

Early copyrighting also predicts subsequent inventive activity. Table 5 relates copyright registrations in 1840–1860 to the log of patents per capita per decade over 1860–1940. This comparison connects the production of written work to a separate record of technological invention during

industrialization. The complete sequence of OLS and IV specifications shows how the relationship changes as geography, education, urbanization, population, and migration are taken into account.

Table 5. Copyrighting and patenting, 1860–1940

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Ordinary least squares</i>					
Log copyright registrations	0.486 (9.30)	0.441 (12.28)	0.400 (10.55)	0.360 (7.47)	0.195 (3.40)
Counties	1,506	1,501	1,499	1,499	1,498
R^2	0.104	0.147	0.179	0.181	0.223
<i>Panel B. Instrumental variables</i>					
Log copyright registrations	1.448 (4.48)	1.416 (4.54)	1.344 (4.51)	2.353 (3.59)	53.89 (0.10)
Counties	1,506	1,501	1,499	1,499	1,498
Kleibergen–Paap F statistic	24	28	22	14	1
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geography controls	No	Yes	Yes	Yes	Yes
Human capital controls	No	No	Yes	Yes	Yes
Urbanization, 1850	No	No	No	Yes	Yes
Remaining social controls	No	No	No	No	Yes

Notes: The dependent variable is the log of patents per capita per decade over 1860–1940. Copyright registrations refer to 1840–1860 and are assigned to authors’ counties of residence in 1850. Only the copyright coefficient is displayed. Column (5) adds the remaining social controls to the specification in column (4). Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

The OLS relationship is positive and statistically significant throughout. The coefficient falls from 0.486 with state effects and the spatial polynomial to 0.195 in the full specification, where the t statistic is 3.40. Thus, copyright activity contains information about later invention beyond baseline literacy, schooling, urbanization, population, and migration. The persistence of this relationship across the control sequence complements the manufacturing results: the geography of early authorship anticipates both industrial specialization and inventive activity.

The first four IV specifications also yield positive, statistically significant coefficients. The estimates range from 1.344 to 2.353, with first-stage F statistics between 14 and 28. Adding the remaining population and migration controls in column (5) reduces the first-stage statistic to 1 and yields an imprecise coefficient of 53.89 with a t statistic of 0.10. The fully controlled OLS estimate and the IV estimates in columns (1)–(4) together provide the clearest evidence of the positive copyright–patenting relationship.

The unit of this relationship is the county. Places with more registered writing in the middle of the nineteenth century subsequently produced more patents relative to their populations over the industrialization period. The two records capture different forms of intellectual activity at different dates. Their association provides a second dimension of the relationship between early authorship and industrial development.

5.3 Urbanization

Table 6 relates early copyrighting to the growth of the urban population share over 1860–1940. Urbanization captures a different dimension of development from manufacturing specialization:

it asks whether counties with more written activity experienced a larger shift of population into urban places. Every specification conditions on the urban share in 1850, with geography, human capital, and the remaining social controls added sequentially.

Table 6. Copyrighting and urban population share growth, 1860–1940

	(1)	(2)	(3)	(4)
<i>Panel A. Ordinary least squares</i>				
Log copyright registrations	0.0654 (4.45)	0.0617 (4.36)	0.0548 (3.92)	0.0169 (1.11)
Urban population share, 1850	−0.503 (−7.98)	−0.504 (−7.66)	−0.482 (−7.51)	−0.712 (−8.02)
Counties	1,472	1,467	1,466	1,465
<i>Panel B. Instrumental variables</i>				
Log copyright registrations	0.328 (2.38)	0.365 (2.45)	0.358 (2.13)	−2.832 (−0.05)
Urban population share, 1850	−1.200 (−3.44)	−1.279 (−3.62)	−1.260 (−3.12)	1.954 (0.04)
Counties	1,472	1,467	1,466	1,465
Kleibergen–Paap F statistic	16	17	13	1
State fixed effects	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes
Geography controls	No	Yes	Yes	Yes
Human capital controls	No	No	Yes	Yes
Urbanization, 1850	Yes	Yes	Yes	Yes
Remaining social controls	No	No	No	Yes

Notes: The outcome is growth in the urban population share between 1860 and 1940. Copyright registrations refer to 1840–1860 and are assigned to authors’ counties of residence in 1850. All columns include baseline urbanization; the final column additionally includes the remaining social controls. Only selected coefficients are displayed. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

Both OLS and IV estimates are positive and statistically significant in the first three specifications. The OLS coefficients range from 0.0548 to 0.0654, and the IV coefficients from 0.328 to 0.365. Counties with more copyrighting experienced larger increases in their urban population shares conditional on initial urbanization, geography, and human capital.

The additional population and migration controls account for a substantial part of this association. In the final column, the OLS coefficient is 0.0169 with a t statistic of 1.11; the IV first-stage statistic falls to 1 and the estimate becomes imprecise. Urban population growth was therefore closely related to the broader settlement characteristics of copyrighting counties. Manufacturing and patenting exhibit the more persistent conditional relationships with early authorship.

5.4 Population density and longer-run outcomes

Figure 5 plots the copyright coefficients for population density from 1860 through 2000. The relationship remains positive throughout the displayed period. The IV series becomes larger over much of the twentieth century, while the OLS series is more stable. Read alongside the manufacturing results, the figure suggests that the local concentration of population persisted beyond the period when early copyrighting was most strongly associated with manufacturing employment shares.

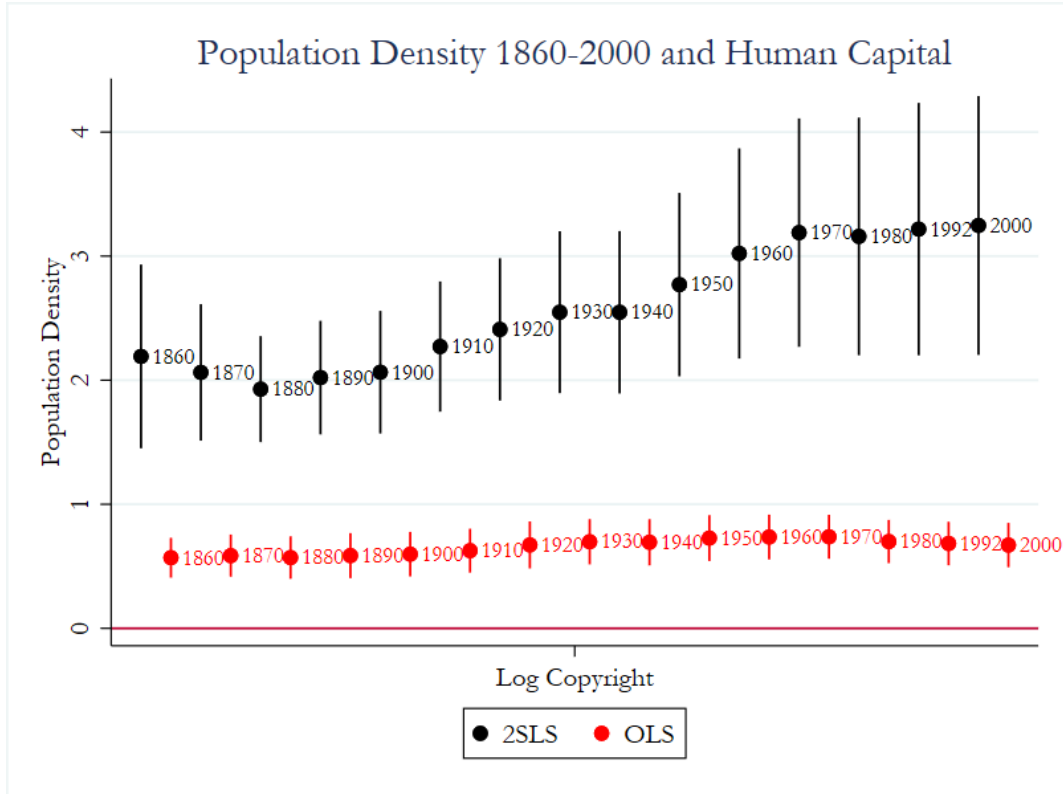


Figure 5. Copyrighting and population density, 1860–2000

Notes: Copyright coefficients and 95% confidence intervals from separate county regressions for the years shown. Black points are 2SLS and red points are OLS. The time-pattern specifications retain geography, human capital, baseline urbanization, state effects, and the spatial polynomial and exclude the remaining social controls.

Appendix Table D1 extends the analysis to income per capita in 2000, using the same sequence of geographic, educational, and settlement controls.

The composition of copyright activity adds another dimension to these results. Appendix C separates academic from non-academic works. Both categories are positively associated with manufacturing in 1860 and 1870; the non-academic coefficient remains positive and statistically significant in 1920. The aggregate relationship consequently extends across more than one type of written work. Appendix D.2 places the copyright measure alongside agricultural diversity in joint specifications, connecting this evidence to the wider study of local conditions and American structural change.

5.5 Interpretation

The results identify a coherent historical pattern. Counties with greater copyright activity in 1840–1860 had larger manufacturing employment shares during industrialization and recorded higher patenting rates over 1860–1940. These relationships remain visible after conditioning on conventional measures of education and initial settlement. The manufacturing association reaches its largest selected-year estimate in 1920, while the continued concentration of population indicates that the geographical pattern extends beyond industrial specialization itself.

Copyright records give this pattern an intellectual dimension. They locate people engaged in producing written work and reveal part of the subject matter on which they wrote. The positive historical associations for academic and non-academic works suggest a broad connection

between authorship and the development of an industrial economy. Together, the evidence establishes early copyrighting as a measure of local intellectual activity with explanatory content beyond literacy, schooling, and population, and links that activity to the subsequent geography of American manufacturing and invention.

6 Conclusion

This paper introduces evidence on the production of written knowledge in the early United States and connects its geography to American industrial development. Copyright title pages preserved by the Library of Congress record a sustained rise in registrations per capita beginning around 1830. Linking authors to the 1850 census locates this intellectual activity at the county level and identifies the people behind it. The resulting data bring written production into the study of human capital during the industrial transition.

Early copyright activity predicts both manufacturing and invention. Counties with more registrations developed larger manufacturing employment shares during the industrial expansion, and the relationship is strongest around the early twentieth century. Copyrighting also predicts patenting over 1860–1940 after accounting for geography, education, population, and migration. Positive IV estimates based on settlers’ origins provide a second source of evidence on these relationships. The composition analysis shows that the manufacturing association extends to non-academic writing, while the longer-run evidence connects early copyright activity to persistent population concentration.

These findings place the growth of intellectual production within the history of American industrialization. Written activity was expanding and geographically concentrated before the industrial surge of the late nineteenth century. Its local distribution helps explain where manufacturing and invention subsequently flourished. Copyright records thus reveal a dimension of human capital that complements literacy and schooling and connects the people producing written knowledge to the economic transformation of their communities.

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A Matching diagnostics and author characteristics

The linkage procedure in Section 2.3 combines extracted names, a state-restricted initial search, phonetic coding, occupation restrictions, and an age restriction. Table A1 reports the number of candidate matches associated with entries in the diagnostic sample. Of the 9,662 entries, 6,927 (71.69%) identify a single census candidate under the search restrictions. The table describes candidate multiplicity at the matching-entry level. The demographic comparison in Table A3 reports the characteristics of 5,162 linked authors.

Table A1. Number of candidate census matches

Candidate matches	Frequency	Percent
1	6927	71.69
2	990	10.25
3	585	6.055
4	364	3.767
5	255	2.639
6	180	1.863
7	133	1.377
8	88	0.911
9	90	0.931
10	50	0.517
Total	9662	100

Notes: The first column reports the number of census candidates associated with a name after applying the matching restrictions. A single candidate denotes a unique match under the stated name, location, occupation, and age restrictions. Sources: Library of Congress copyright records and the 1850 Census of Population.

Table A2 lists the census occupations admitted to the candidate pool and their frequencies. The set includes professional and technical occupations, including teachers, clergy, physicians, and lawyers. This occupational screen narrows the census search to people whose reported work is consistent with authorship. The same eligible pool supplies the comparison group for the demographic analysis below.

Table A2. Occupations eligible for author-to-census matching

Occupation	Frequency	Percent
Accountants and auditors	1215	0.802
Architects	572	0.377
Artists and art teachers	2238	1.477
Authors	96	0.0634
Chemists	436	0.288
Clergymen	27698	18.28
College presidents and deans	79	0.0521
Agricultural sciences	1	0.000660
Chemistry	21	0.0139
Engineering	2	0.00132
Geology and geophysics	2	0.00132
Mathematics	32	0.0211
Medical sciences	11	0.00726
Physics	3	0.00198
Natural science (n.e.c.)	8	0.00528
Social sciences (n.e.c.)	12	0.00792
Nonscientific subjects	160	0.106
Subject not specified	539	0.356
Dentists	3029	1.999

Continued on next page

Table A2 (continued)

Occupation	Frequency	Percent
Designers	113	0.0746
Draftsmen	104	0.0686
Editors and reporters	1526	1.007
Engineers, civil	812	0.536
Engineers, electrical	1	0.000660
Engineers, industrial	3	0.00198
Engineers, mechanical	49	0.0323
Engineers, mining	5	0.00330
Engineers (n.e.c.)	37	0.0244
Lawyers and judges	25129	16.58
Musicians and music teachers	3808	2.513
Nurses, professional	6	0.00396
Agricultural scientists	7	0.00462
Biological scientists	55	0.0363
Geologists and geophysicists	27	0.0178
Physicists	13	0.00858
Miscellaneous natural scientists	13	0.00858
Pharmacists	6209	4.098
Physicians and surgeons	44721	29.51
Religious workers	406	0.268
Statisticians and actuaries	4	0.00264
Teachers (n.e.c.)	31217	20.60
Technicians, medical and dental	3	0.00198
Technicians, testing	23	0.0152
Technicians (n.e.c.)	3	0.00198
Therapists and healers (n.e.c.)	370	0.244
Veterinarians	142	0.0937
Professional, technical and kindred workers (n.e.c.)	371	0.245
Opticians and lens grinders and polishers	200	0.132
Total	151531	100

Notes: Census occupational categories and frequencies reproduce the set eligible for matching. “N.e.c.” means not elsewhere classified. Frequencies describe the census population eligible for matching. Source: 1850 Census of Population.

Table A3 compares matched authors with non-authors in the eligible occupational pool. Authors are older on average and have higher reported real estate values. The foreign-born share is 15.8% among authors and 11.8% among eligible non-authors. The comparison therefore identifies demographic and economic differences between authors and other people working in the same set of eligible occupations.

Table A3. Characteristics of linked authors and other eligible census individuals

	Authors		Non-authors		Difference
	Mean	SD	Mean	SD	<i>t</i> statistic
White	0.990	0.098	0.995	0.073	3.105
Real estate value	2395.759	11704.767	1679.085	10034.188	−4.343
Foreign-born	0.158	0.364	0.118	0.323	−7.658
Age	38.810	12.924	35.834	12.731	−16.271
Observations	5162		146369		151531

Notes: Non-authors are individuals in the eligible occupations in Table A2 who are not linked to copyright registrations. White and foreign-born are binary indicators; real estate value is measured in 1850 dollars. The final column reports difference-in-means *t* statistics, with non-authors first in the comparison. Source: 1850 Census of Population linked to Library of Congress copyright records.

B Variables and data sources

Copyright data are constructed from Library of Congress title-page transcriptions for 1790–1870. County copyrighting aggregates the registrations in 1840–1860 assigned to a matched author’s residence in the 1850 census. The national time series, the county registration measure, and the origin-specific author counts therefore use different units. The instrument uses authors divided by census residents from an origin, as defined in equations (2)–(3).

Historical economic outcomes and geographic variables are drawn primarily from Fiszbein (2022). Census-based urbanization, educational variables, and church counts are drawn from Haines and ICPSR (2010). Table B1 groups the explanatory variables. Baseline urbanization enters separately in the sequential specifications, allowing its role to be distinguished from that of population and migration.

The manufacturing outcome is the employment share of manufacturing. The patenting regressions use the log of patents per capita per decade during 1860–1940. Urban population-share growth is the difference between the urban shares in 1940 and 1860. Income regressions use log income per capita in 2000. The population-density figure displays year-specific estimates for 1860–2000.

Table B1. Control groups used in the county regressions

Group	Variables
Geography	Potential yields; average rainfall; temperature; distance to oceans or Great Lakes; terrain ruggedness.
Human capital	Literacy rate; school enrollment rate.
Social characteristics	Log population; urbanization; fraction of foreign-born residents; internal migrants; population density.

Notes: Human-capital and social characteristics are measured in 1850. Specifications also include state fixed effects and a third-degree polynomial in latitude and longitude. In the sequence of nested specifications, urbanization enters before the remaining social characteristics. Sources: Fiszbein (2022) and Haines and ICPSR (2010).

C Title-page classification

C.1 Procedure and interpretation

The subject matter recorded on title pages provides a basis for separating different forms of written production. The binary classification used in the county map and supplementary regressions distinguishes academic from non-academic writing. In this application, academic writing means work directed toward transmitting or advancing knowledge that is not primarily religious in content. It includes instructional materials and is broader than scientific research. A work written by a member of the clergy, including a teaching manual, can therefore fall in the academic category.

The procedure first ranks title pages by the combined relative frequency of the terms in Table C1. The top and bottom deciles supply positive and negative labels, respectively. A logistic classifier fitted to word counts in these labeled subsets is then used to classify the remaining entries. Writing x_j for the vector of word counts on title page j , the model can be expressed as

$$\Pr(D_j = 1 \mid x_j) = \Lambda(\alpha + x_j' b), \quad \Lambda(v) = \frac{\exp(v)}{1 + \exp(v)}, \quad (\text{C1})$$

where D_j is the keyword-derived binary label in the training subset. The classifier extends the keyword-based labels to the full set of entries, producing a consistent division of title-page text into the two categories.

Table C1. Terms used to initialize the academic classification

textbook	reader	instruction
principal	grammar	school
Sundayschool	Sabbath school	manual
teacher	university	professor

Notes: Terms are reproduced from the keyword list. The list is used to rank title pages before selecting the top and bottom deciles for classification.

A separate set of keyword flags identifies narrower topics. These flags allow overlapping subject categories; the binary classifier assigns every classified work to one of two groups. The main regressions use total copyright activity, and the composition analysis enters academic and non-academic registrations separately.

C.2 Manufacturing by type of written activity

Table C2 separates the manufacturing relationship by the content of written production. Both academic and non-academic registrations have positive, statistically significant coefficients in 1860 and 1870. In 1920, the coefficient on non-academic writing remains positive and significant at 0.0105 ($t = 3.25$), while the academic coefficient is close to zero. The historical association between copyrighting and manufacturing therefore extends across a broad range of written work. Its persistence for the non-academic category connects industrial development to intellectual activity beyond formal academic subjects.

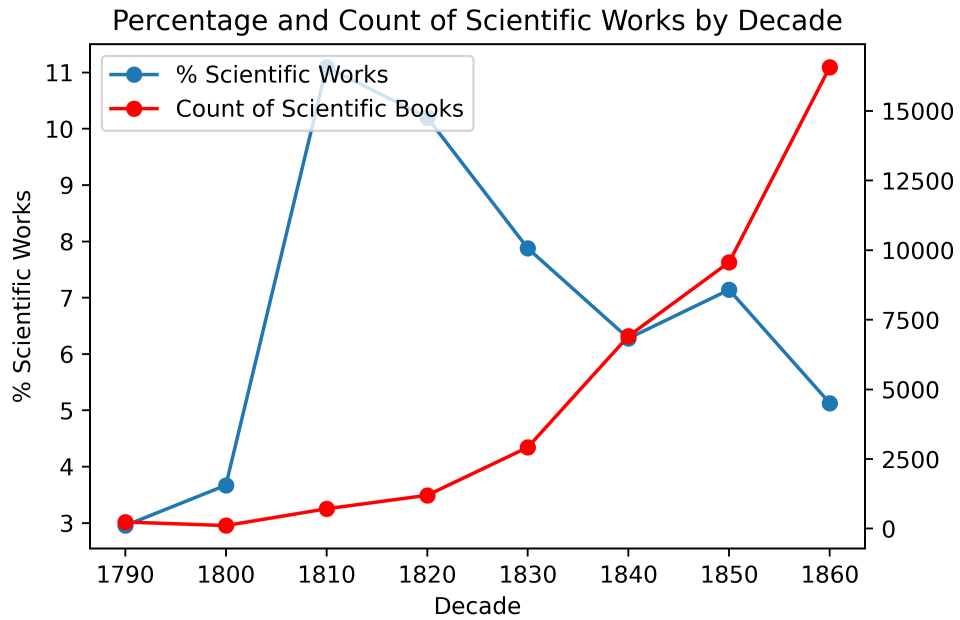
Table C2. Academic and non-academic copyrighting and manufacturing employment

Outcome year	(1) 1860	(2) 1870	(3) 1920	(4) 1970	(5) 2000
Log academic registrations	0.00899 (3.34)	0.00785 (2.10)	-0.00307 (-0.43)	-0.00264 (-0.45)	0.00708 (1.65)
Log non-academic registrations	0.00670 (4.97)	0.00873 (3.50)	0.0105 (3.25)	-0.00254 (-0.80)	-0.00246 (-0.54)
Observations	1504	1504	1504	1504	1504
R^2	0.295	0.195	0.203	0.221	0.130
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude-longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes	Yes
Human-capital controls	Yes	Yes	Yes	Yes	Yes
Social controls	Yes	Yes	Yes	Yes	Yes

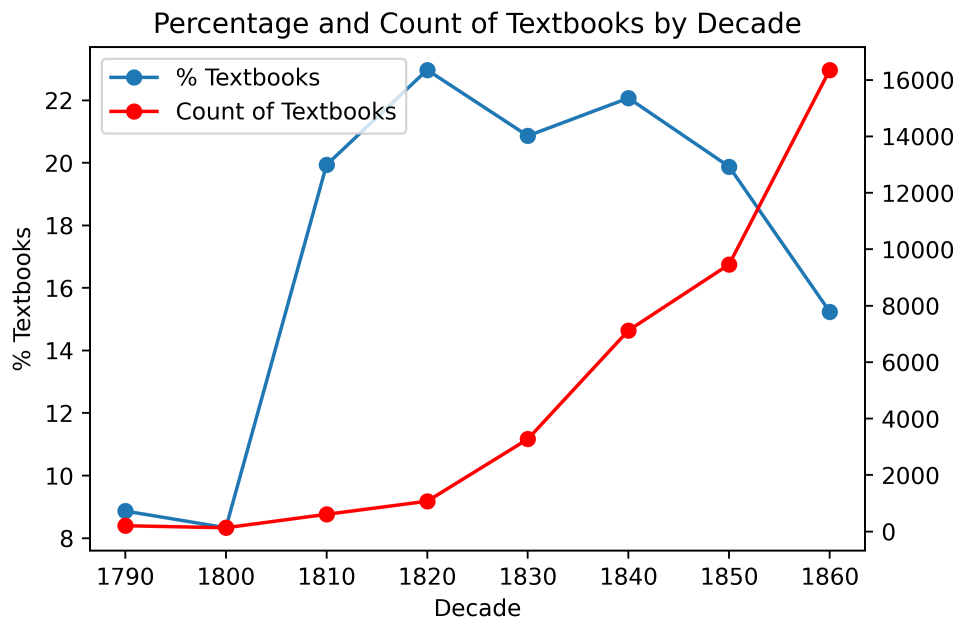
Notes: OLS estimates. The dependent variable is the manufacturing share of employment in the indicated year. The regressors distinguish academic and non-academic registrations over 1840–1860, using the title-based classification described in the appendix. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

C.3 Supplementary subject-matter series

Figure C1 shows the series for scientific and textbook-oriented writing. Each image shows a count and a relative frequency, tracing both the expansion of these categories and their share in recorded written production.



(a) Scientific titles



(b) Textbook-oriented titles

Figure C1. Supplementary classification series

Notes: Count and relative-frequency series. The panels use different subject-specific keyword labels. The manufacturing regressions use the separate binary academic/non-academic classification. Source: author's classification of copyright title-page text.

D Supplementary development results

D.1 Income per capita in 2000

Table D1 relates early copyright activity to income per capita in 2000. The OLS and IV coefficients are positive and statistically significant through column (4), which includes geography, human capital, and baseline urbanization. Adding the population and migration controls in column (5) reduces the OLS coefficient to 0.0135 ($t = 1.04$). The IV first-stage statistic falls to one in that specification, leaving the corresponding income coefficient imprecisely estimated. The specification sequence shows how the long-run income association varies with the controls for initial settlement.

Table D1. Copyrighting and income per capita in 2000

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Ordinary least squares</i>					
Log copyright registrations	0.0861 (7.09)	0.0704 (5.43)	0.0629 (5.13)	0.0500 (4.42)	0.0135 (1.04)
R^2	0.0877	0.155	0.178	0.183	0.232
<i>Panel B. Instrumental variables</i>					
Log copyright registrations	0.417 (6.12)	0.421 (5.82)	0.406 (5.14)	0.782 (3.24)	7.988 (0.12)
Kleibergen–Paap F	24	28	22	13	1
Observations (both panels)	1512	1507	1505	1505	1504
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geographic controls	No	Yes	Yes	Yes	Yes
Human-capital controls	No	No	Yes	Yes	Yes
Baseline urbanization	No	No	No	Yes	Yes
Remaining social controls	No	No	No	No	Yes

Notes: The dependent variable is log income per capita in 2000. Copyright registrations cover 1840–1860. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff. Column (5) adds the full social-control group, including log population, to the controls in column (4). The control groups are listed in Table B1.

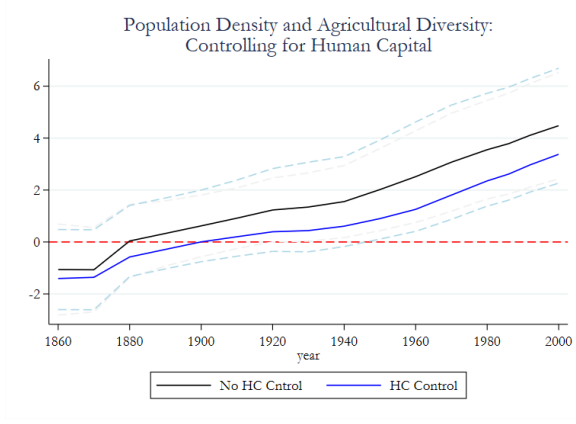
D.2 Agricultural diversity

Fiszbein (2022) studies the relationship between agricultural diversity and American structural change. The supplementary joint specification is

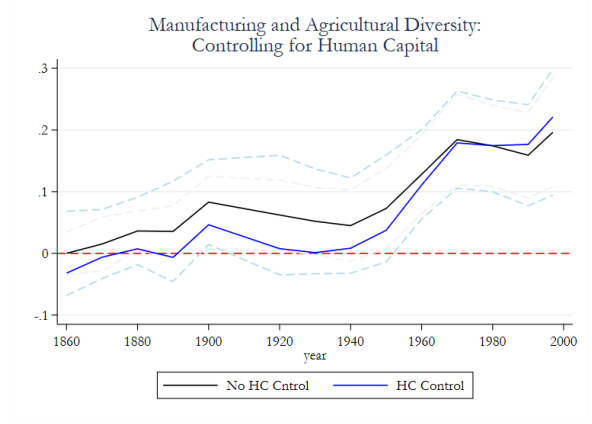
$$y_{it} = \alpha_{s(i),t} + \beta_t k_i + \gamma_t A_i + X_i' \delta_t + \varepsilon_{it}, \quad (\text{D1})$$

where A_i denotes agricultural diversity in 1860. Copyrighting is instrumented with the migration-composition index and agricultural diversity with the instrument used in Fiszbein (2022); this exercise retains that analysis’s baseline controls. Figure D1 displays the coefficient paths.

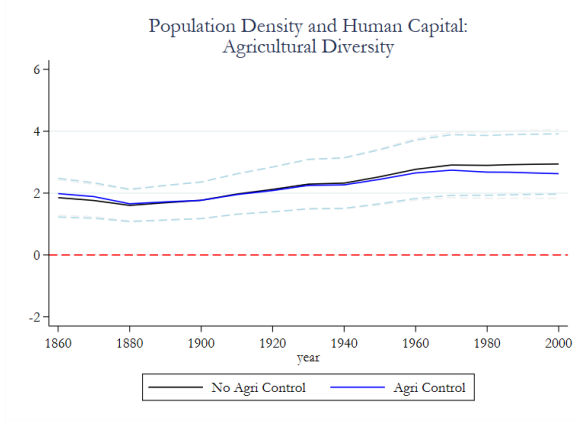
Copyrighting retains positive historical associations when agricultural diversity enters the same specification. The estimated agricultural-diversity relationship also changes when copyrighting is included. The joint analysis therefore shows that early intellectual production contains information about local development conditional on agricultural diversity. Its IV interpretation maintains the exclusion restrictions for both instruments.



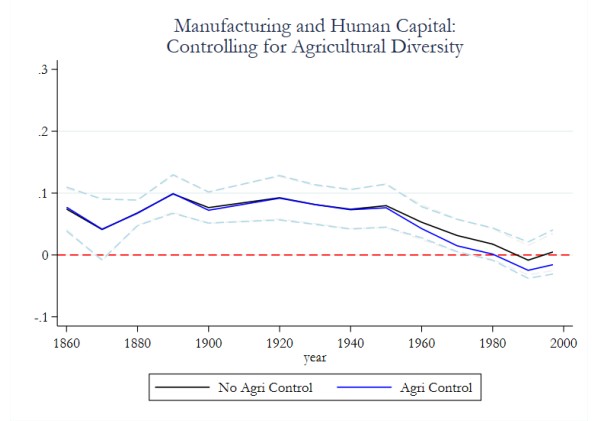
(a) Diversity: population density



(b) Diversity: manufacturing share



(c) Copyrighting: population density



(d) Copyrighting: manufacturing share

Figure D1. Copyrighting and agricultural diversity in joint specifications

Notes: Reported IV coefficient paths from the supplementary joint analysis. Panels (a) and (b) show agricultural-diversity coefficients conditional on copyrighting; panels (c) and (d) show copyright coefficients conditional on agricultural diversity. The agricultural-diversity instrument follows Fiszbein (2022). These results are distinct from the baseline specifications in the main text.

D.3 Religious composition and copyrighting

The county data also permit a descriptive comparison between copyrighting and the denominational composition of local churches. Following Montalvo and Reynal-Querol (2003), let n_{ij} denote the number of churches belonging to denomination j in county i , let $N_i = \sum_j n_{ij}$, and define the denomination's share as $\pi_{ij} = n_{ij}/N_i$. Fractionalization and polarization are

$$\text{FRAC}_i = 1 - \sum_{j=1}^J \pi_{ij}^2, \quad (\text{D2})$$

$$\text{POL}_i = 1 - \sum_{j=1}^J \left(\frac{0.5 - \pi_{ij}}{0.5} \right)^2 \pi_{ij}. \quad (\text{D3})$$

Fractionalization measures the probability that two churches drawn at random belong to different denominations. Polarization is greatest when two denominations each account for half of the churches. The indices summarize the county distribution of religious organizations. Church counts are drawn from the historical county data assembled by Haines and ICPSR (2010).

Table D2 estimates their partial correlations with log copyright registrations over 1840–1860, including both indices in the same regression. Fractionalization is positively associated with copyrighting, whereas polarization is negatively associated with it. Both relationships remain statistically significant as the control set expands. With the full set of controls, the respective coefficients are 0.268 and -0.215 .

Measured in 1850, the church indices describe the religious composition of counties during the registration window. The estimates show that written production was more prevalent where churches were spread across denominations and less prevalent where the denominational distribution was polarized.

Table D2. Religious composition and county copyrighting

	(1)	(2)	(3)	(4)	(5)
Fractionalization	0.582 (7.30)	0.558 (8.05)	0.523 (7.92)	0.523 (7.92)	0.268 (5.59)
Polarization	-0.505 (-6.38)	-0.476 (-6.73)	-0.445 (-6.23)	-0.445 (-6.23)	-0.215 (-5.14)
Observations	1511	1506	1505	1505	1504
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geographic controls	No	Yes	Yes	Yes	Yes
Human-capital controls	No	No	Yes	Yes	Yes
Social controls	No	No	No	No	Yes

Notes: OLS estimates. The dependent variable is log county copyright registrations over 1840–1860. Both religious-composition indices enter each specification. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff. Control groups are defined in Table B1.